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Batch 15

Algorithm

1. Start.
2. Read the number of rows and columns.
3. Validate that rows and columns are positive.
4. Read all elements of the 2D matrix.
5. Repeatedly display the menu:
 - 1 – Sum of all matrix elements
 - 2 – Sum of each row and store the sums in an array
 - 3 – Transpose of the matrix
 - 4 – Exit
6. Read the user's choice.
7. Use a **switch** statement:
 - **Choice 1:** Initialize **sum** = 0. Traverse the entire matrix using nested loops and add every element to **sum**. Display the sum.
 - **Choice 2:** For each row, initialize its sum to 0, add all elements in that row, and store the result in the row-sum array. Display each row sum and the resulting array.
 - **Choice 3:** Display the transpose by interchanging the row and column indices while printing.
 - **Choice 4:** Exit the program.
 - **Default:** Display an invalid-choice message.
8. Repeat until choice 4 is selected.
9. Stop.

Test-Case Table

Test Case	Input	Operation	Expected Output
1	Matrix = { { 1, 2, 3 }, { 4, 5, 6 } }	Sum of all elements	21
2	Matrix = { { 1, 2, 3 }, { 4, 5, 6 } }	Row sums	[6, 15]
3	Matrix = { { 1, 2, 3 }, { 4, 5, 6 } }	Transpose	{ { 1, 4 }, { 2, 5 }, { 3, 6 } }
4	Matrix = { { 5 } }	Sum	5
5	Matrix = { { -1, 2 }, { 3, -4 } }	Sum	0
6	Rows = 0 or Columns = -2	Invalid dimensions	Ask for dimensions again
7	Choice = 7	Invalid menu choice	Invalid choice!

8	Choice = 4	Exit	Program terminates
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C Program

```
#include <stdio.h>

int main()
{
    int rows, cols, i, j, choice, sum;

    /* Input and validation of dimensions */
    do
    {
        printf("Enter the number of rows and columns: ");
        scanf("%d %d", &rows, &cols);

        if (rows <= 0 || cols <= 0)
            printf("Rows and columns must be positive. Try again.\n");
    }
    while (rows <= 0 || cols <= 0);

    /* Variable Length Arrays */
    int matrix[rows][cols];
    int rowSum[rows];

    /* Input matrix */
    printf("\nEnter the matrix elements:\n");

    for (i = 0; i < rows; i++)
    {
        for (j = 0; j < cols; j++)
        {
            scanf("%d", &matrix[i][j]);
        }
    }

    do
    {
        printf("\nMENU\n");
        printf("1. Find sum of all elements\n");
        printf("2. Find sum of each row\n");
        printf("3. Find transpose of matrix\n");
```

```

printf("4. Exit\n");

printf("\nEnter your choice: ");
scanf("%d", &choice);

switch (choice)
{
    case 1:
        sum = 0;

        for (i = 0; i < rows; i++)
        {
            for (j = 0; j < cols; j++)
            {
                sum = sum + matrix[i][j];
            }
        }

        printf("Sum of all elements = %d\n", sum);
        break;

    case 2:
        printf("\nRow sums:\n");

        for (i = 0; i < rows; i++)
        {
            rowSum[i] = 0;

            for (j = 0; j < cols; j++)
            {
                rowSum[i] = rowSum[i] + matrix[i][j];
            }

            printf("Sum of row %d = %d\n",
                i + 1, rowSum[i]);
        }

        printf("\nThe row-sum array contains: [");

        for (i = 0; i < rows; i++)
        {
            printf("%d", rowSum[i]);

            if (i < rows - 1)
                printf(", ");
        }
    }
}

```

```

        }

        printf("]\n");
        break;

    case 3:
        printf("\nTranspose of the matrix:\n");

        for (j = 0; j < cols; j++)
        {
            for (i = 0; i < rows; i++)
            {
                printf("%d\t", matrix[i][j]);
            }
            printf("\n");
        }
        break;

    case 4:
        printf("Exiting the program...\n");
        break;

    default:
        printf("Invalid choice! Please enter 1 to
4.\n");
    }

}

while (choice != 4);

return 0;
}

```

Execution

```

Last login: Thu Sep 24 08:08:03 on console
bhavya@bhavyas-MacBook-Air ~ % nano day31.cpp -o day31
bhavya@bhavyas-MacBook-Air ~ % ./day31
zsh: no such file or directory: ./day31
bhavya@bhavyas-MacBook-Air ~ % clang day31.cpp -o day31
bhavya@bhavyas-MacBook-Air ~ % ./day31
Enter the number of rows and columns: 2 3

Enter the matrix elements:
1 2 3
4 5 6

MENU
1. Find sum of all elements
2. Find sum of each row
3. Find transpose of matrix
4. Exit

Enter your choice: 1
Sum of all elements = 21

MENU
1. Find sum of all elements
2. Find sum of each row
3. Find transpose of matrix
4. Exit

Enter your choice: 2

Row sums:
Sum of row 1 = 6
Sum of row 2 = 15

The row-sum array contains: [6, 15]

MENU
1. Find sum of all elements
2. Find sum of each row
3. Find transpose of matrix
4. Exit

Enter your choice: 3

Transpose of the matrix:
1      4
2      5
3      6

MENU
1. Find sum of all elements
2. Find sum of each row
3. Find transpose of matrix
4. Exit

Enter your choice: 4
Exiting the program...
bhavya@bhavyas-MacBook-Air ~ % █

```